

Two Player Games

QuantProf revision · each problem: the idea in plain words with a playable diagram, then a numbered proof.

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| 0 · Ground rules | 3 · Winning / losing states |
| 1 · Symmetry | 4 · Endgame analysis |
| 2 · Strategy stealing | 5 · Ad-hoc |

0 · Ground rules

1. **Setting:** two players alternate, with the same moves available to both. The player with no legal move loses. The game is finite, so there are no draws.
2. **L-position** (losing for the player to move): every move goes to a W-position. A position with no moves is L.
3. **W-position:** some move goes to an L-position.
4. In a finite game every position is exactly one of W or L. (induction backwards from the end)

So to prove a player wins, give them a way to always move into L.

1 · Symmetry

Make the position symmetric (you move first) or keep it symmetric (you move second). Then copy the

opponent's move on the mirror image. If they had a move, you have one too, so you are never the one stuck.

Removing petals

A flower has 11 petals arranged in a circle. Alice and Bob take turns, Alice first. On a turn a player tears off either one petal, or two petals that are next to each other (with no gap between them). The player who cannot make a move loses.

Question: which player can always win?

Key idea

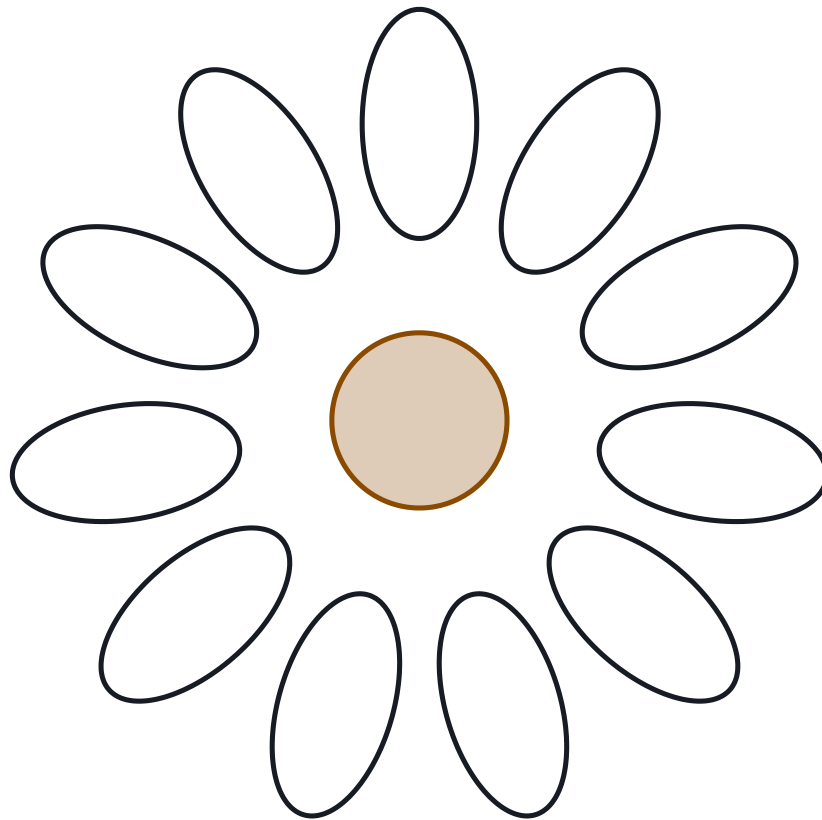
Alice's first move turns the circle into a line. Bob cuts that line into two equal halves and then mirrors Alice.

Picture

Bob turns the game into two identical halves, then copies Alice.

Alice's first move breaks the circle into a line. Bob removes the middle of that line, leaving two equal lines of 4. From then on, whatever Alice takes from one line, Bob takes the same petals from the other. Alice runs out of moves first.

You are Alice. Click one petal, or two neighbouring petals, then press Take. Bob answers with the strategy from the proof.

[Take](#)[Restart](#)

Your move.

Solution

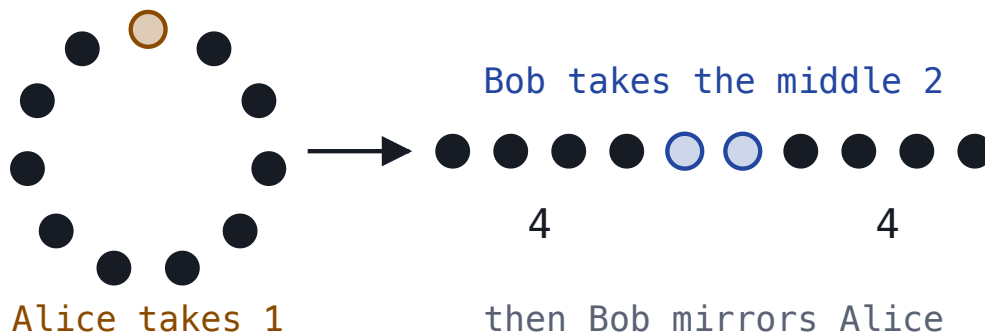
Setup. 11 petals in a circle. A move removes 1 petal or 2 *adjacent* petals (neighbours with no gap between them). The player who can't move loses. Alice moves first.

Claim. Bob wins.

1. After Alice's first move, the remaining petals form a single line of 10 (if she took 1) or 9 (if she took 2).

Why: removing petals from a circle breaks it at that spot; the petals on either side of a gap are not adjacent.

2. Bob removes the middle 2 petals of the line of 10, or the middle 1 petal of the line of 9. Either way, two separate lines of 4 petals remain.



Why: $10 - 2 = 8 = 4 + 4$ and $9 - 1 = 8 = 4 + 4$, with the gap in the middle.

3. From then on, pair petal j of one line with petal j of the other. Whatever Alice removes from one line (1 petal or 2 adjacent ones), Bob removes the matching petals from the other line.

Why: this is Bob's strategy.

4. Bob's reply is always legal, and after it the two lines are identical again.

Why: before Alice's move the lines were identical, so the matching petals are present and adjacent in the other line. No move can use petals from both lines, because they are not adjacent.

5. The game is finite and Bob always has a reply, so Alice is the one who gets stuck.

Why: each move removes at least one petal.

Conclusion. Bob splits the line into two equal halves and mirrors Alice. ■

△ Course says "Bob can always leave two identical rows" but not how. Step 2 is the how.

Chords

20 points are marked on a circle. Alice and Bob take turns, Alice first. A move draws a straight segment joining two of the points; it must not cross any segment already drawn (segments may share an endpoint). The player who cannot move loses.

Question: which player can always win?

Key idea

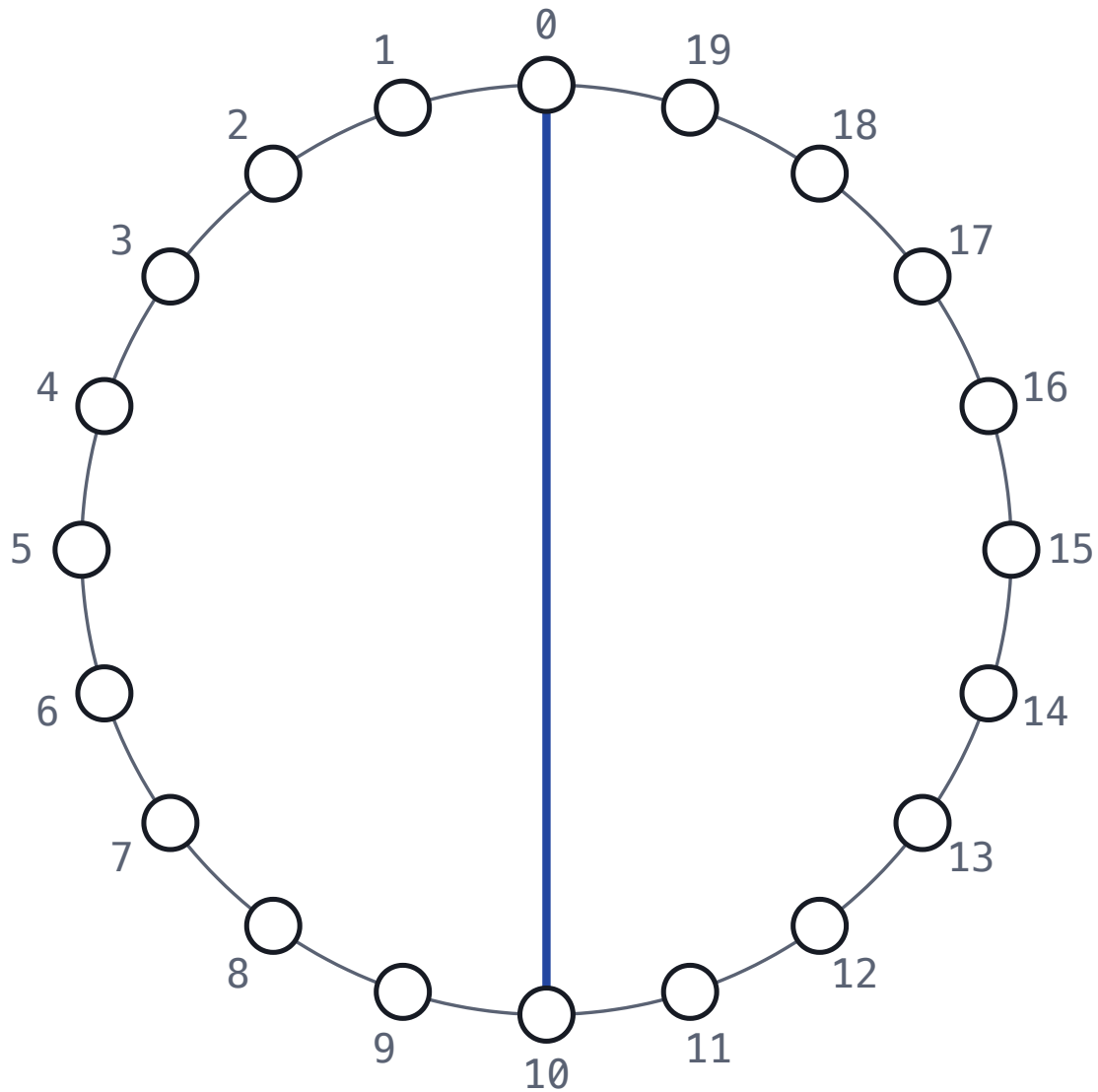
Alice draws a "diameter" with 9 points on each side, then answers every move with its mirror image across that diameter.

Picture

Alice draws a diameter, then copies Bob in the mirror.

The diameter splits the points into two equal halves. Every chord Bob draws stays in one half (it can't cross the diameter), and its mirror image in the other half is always free and legal. Bob runs out first.

You are Bob. Alice has drawn the diameter 0–10. Click two points to draw a chord; Alice replies with its mirror image across the diameter.


[Restart](#)

Your move: click two points.

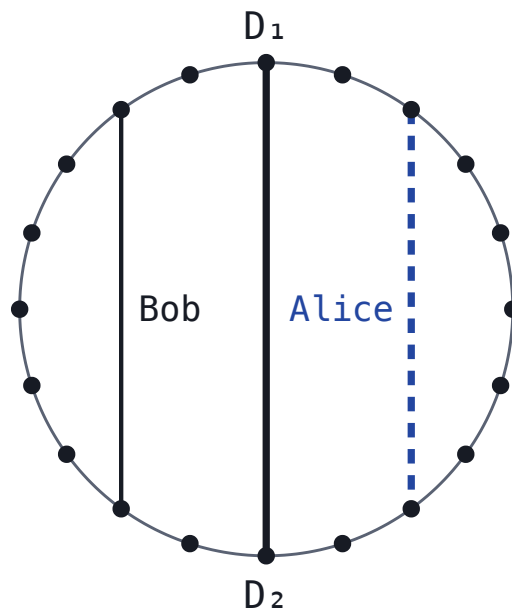
Solution

Setup. 20 points on a circle. A move draws a segment between two points that crosses no segment already drawn (sharing an endpoint is allowed). The player who can't move loses. Alice moves first.

Claim. Alice wins.

1. Only the cyclic order of the points matters, so place them evenly. Alice joins two opposite points D_1, D_2 , leaving 9 points on each side. Let σ be

the reflection in the line D_1D_2 ; it swaps the two sides.



Why: crossing depends only on the order of endpoints around the circle.

2. Alice's rule: when Bob draws a chord c , she draws $\sigma(c)$. Invariant: after each of Alice's moves, the drawing is symmetric under σ .

Why: true after her first move, and each reply adds the mirror of Bob's chord.

3. Bob's chord c can't cross D_1D_2 , so it lies in one closed half, and $\sigma(c)$ lies in the other.

Why: D_1D_2 is already drawn.

4. $\sigma(c)$ is a legal move.

Not already drawn: otherwise $c = \sigma(\sigma(c))$ was drawn too, by the invariant.

Doesn't cross c : they lie in opposite halves and can meet only on D_1D_2 , at shared endpoints.

Doesn't cross an older chord e : otherwise c would cross $\sigma(e)$, which is also drawn by the invariant.

Why: reflection preserves crossings.

5. So Alice always has a move. There are finitely many chords, so Bob gets stuck.

Why: steps 2–4.

Conclusion. Alice draws a diameter and mirrors Bob across it. ■

△ Course says “reply symmetrically” without naming the symmetry (reflection in the first chord) or checking that the reply is legal (step 4).

2 · Strategy stealing

Proves that the first player wins without finding the strategy. Find a “waiting” move w that the opponent can never use against you.

1. If w leads to an L-position, play w .
2. Otherwise the opponent has a winning reply x after w . Show that you could have reached the same position by playing x yourself first. Then play x .

Needs a finite game with no draws.

Game of Reduction

The numbers 1000, 2000 and 2024 are written on a board. Alice and Bob take turns, Alice first. A move replaces the **largest** number with any smaller

positive whole number, as long as the three numbers stay different. The player who cannot move loses.

Question: which player wins?

Key idea

2024 $\rightarrow$ 2001 is a waiting move: Bob then has to replace 2001 with some v , and Alice could have replaced 2024 with v directly.

Picture

Alice has a "waiting move" that Bob can never exploit.

Either some position $\{1000, 2000, v\}$ is losing for the player to move, and Alice jumps there directly. Or all of them are winning, and Alice plays 2024 $\rightarrow$ 2001: Bob must then replace 2001 by some $v \leq 1999$, and he himself hands Alice one of those winning positions.

Alice wins either way, without knowing which case is true.

A small version you can solve exactly: numbers $a < b < c$. Alice either moves $c \rightarrow v$ directly, or plays the waiting move $c \rightarrow b + 1$, after which Bob must move $b + 1 \rightarrow v$ himself.

a  5 b  10 c  13

Positions $S(v) = \{5, 10, v\}$ that are losing for the player to move: none.

Case B: every $S(v)$ is winning, so Alice plays the waiting move $13 \rightarrow 11$; Bob must then create some $S(v)$ and hand Alice a winning position.

Exact check: Alice **wins** from $\{5, 10, 13\}$.

Solution

Setup. The numbers $\{1000, 2000, 2024\}$. A move replaces the largest number with a smaller positive integer, keeping all three numbers distinct. The player who can't move loses. Alice moves first.

Claim. Alice wins.

1. The sum strictly decreases, so the game is finite, and every position is W or L.

Why: ground rules.

2. For $v \in \{1, \dots, 1999\}$, $v \neq 1000$, let $S(v) = \{1000, 2000, v\}$. Alice can reach any $S(v)$ directly by moving $2024 \rightarrow v$.

Why: $v < 2024$ and distinct from 1000 and 2000.

3. Case A: some $S(v)$ is L. Alice moves there, Bob faces an L-position, and Alice wins.

Why: definition of L.

4. Case B: every $S(v)$ is W. Alice moves $2024 \rightarrow 2001$ (legal). Now the largest number is 2001, and Bob must replace it with some $v < 2001$, $v \notin \{1000, 2000\}$, i.e. $v \leq 1999$. That position is $S(v)$, which is W for the next player, Alice.

Why: Bob has no other kind of move.

Alice: $2024 \rightarrow 2001$ (waiting move)

Bob: $2001 \rightarrow v$

same as Alice playing $2024 \rightarrow v$ herself

either some $\{1000, 2000, v\}$ is L: Alice goes there,
or all are W: the waiting move hands Bob a W for Alice

5. In both cases Alice wins.

Why: the two cases cover everything.

Conclusion. The waiting move $2024 \rightarrow 2001$ steals any winning reply from Bob. ■

Game of primes

Alice starts by saying 2. Then Bob and Alice take turns saying prime numbers.

Rule: if the last number said was n , and p is the next prime after n , the next player must say a prime m with $n < m \leq n + 2(p - n)$: at most twice as far past n as the next prime is.

Whoever says 4567 (which is prime) wins.

Question: which player wins?

Key idea

Bob is forced to say 3. From 3, Alice can say 5 or 7, and saying 5 forces Bob to say 7. So Alice chooses who says 7.

Picture

The start is forced, then Alice gets to choose who says 7.

From 2 the only legal move is 3 (Bob). From 3, Alice may say 5 or 7, and from 5 the only legal move is 7. So Alice can say 7 herself, or make Bob say it. Whichever of those is good for her, she picks it.

You are Bob. Alice said 2. Choose a legal prime each turn; Alice plays perfectly (computed backwards from 4567).

3

Bob plays randomly to the end

Restart

A: 2

Last number 2; next prime 3; you may say a prime in $(2, 4]$.

Solution

Setup. Alice says 2. From the last number n , with p the next prime after n , the next player may say any prime m with $n < m \leq n + 2(p - n)$. Whoever says 4567 (a prime) wins.

Claim. Alice wins.

1. Jumping past 4567 is never better than saying 4567: whenever a jump past it is legal, the allowed range is an interval that contains $p \leq 4567$ and a number above 4567, so it contains 4567 itself. Treat jumping as forbidden; then the game is finite and every position is W or L.

Why: the allowed range is an interval starting just above n .

2. From 2: $p = 3$, so the range is $\leq 2 + 2 = 4$, and the only prime is 3. Bob must say 3.

Why: the rule.

3. From 3: $p = 5$, so the range is $\leq 3 + 4 = 7$. Alice may say 5 or 7.

Why: the primes in $(3, 7]$.

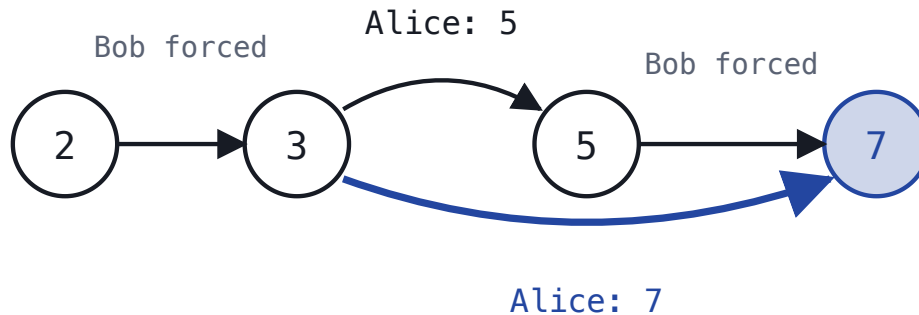
4. From 5: $p = 7$, so the range is $\leq 5 + 4 = 9$, and the only prime is 7. The reply to 5 must be 7.

Why: the primes in $(5, 9]$.

5. The position right after someone says 7 is either W or L for the next player. If it is L, saying 7

is winning, and Alice says 7. If it is W, Alice says 5, and Bob is forced to say 7, handing Alice a W-position.

Why: steps 3 and 4.



Alice picks who says 7 $\Rightarrow$ she wins

Conclusion. Alice controls who says 7, so she wins.
(Checked by brute force.) ■

△ Course ignores the possibility of jumping past 4567 (the game could run forever, a draw). Step 1 closes that gap.

Divisors III

The numbers 1 to 2025 are written on a board. Alice and Bob take turns, Alice first. A move: choose a number still on the board and erase it together with all of its divisors that are still on the board. The player who cannot move (because the board is empty) loses.

Question: which player wins?

Key idea

Erasing just 1 is the waiting move, because 1 divides everything.

Picture

Erasing just the number 1 is a waiting move.

If that hands Bob a losing position, great. If not, Bob has a winning reply "erase x". But erasing x also erases 1 (1 divides everything), so Alice could have erased x herself on move one and reached the very same board.

The same game on $1..n$, solved exactly by computer. Alice's winning first move is shown. Whatever it is, the stealing argument says one exists.

n  10

1..10: Alice **wins**. Her winning first moves: **erase 4**, **erase 6** (erasing 1 is not one of them, so Bob would have a winning reply x, and Alice steals it by erasing x first).

Solution

Setup. The numbers 1–2025 are on the board. A move erases an available number together with all its divisors that are still on the board. The player who can't move loses. Alice moves first.

Claim. Alice wins.

1. The game is finite with no draws, so every position is W or L.

Why: each move erases at least one number.

2. If the position after erasing just 1 is L, Alice erases 1 and wins.

Why: she hands Bob an L-position.

3. Otherwise that position is W, so Bob has a winning reply there: erase some $x > 1$, reaching an L-position. That position has erased exactly 1 and the divisors of x .

Why: definition of W.

4. Alice can reach the same position in one move by erasing x herself: that erases x and all its divisors, including 1.

Why: $1 \mid x$.

5. Either way Alice moves into an L-position and wins.

Why: steps 2–4.

Conclusion. Erasing 1 is a waiting move that Bob can never use against Alice. ■

3 • Winning / losing states

Guess the L-positions from small cases, then prove two things: (i) every move from L goes to W; (ii) from every W there is a move to L.

The small cases come from a rec:

1. **Level:** position n ; $\text{win}(n)$ = the player to move wins.
2. **Choice:** every legal move $n \rightarrow m$.
3. **Check:** no legal move $\rightarrow$ lose.

4. **Move:** $\text{win}(n) = \text{some move reaches a losing } m, \text{ i.e. } \exists m : \neg \text{win}(m).$

300 marbles

A pile has 300 marbles. Alice and Bob take turns, Alice first. When the pile has n marbles, a move removes at least 1 and at most $\lfloor n/2 \rfloor$ of them (at most half the pile). The player who cannot move loses.

Question: which player wins, and what should the winner do first?

Picture

The losing piles are 1, 3, 7, 15, 31, ..., one less than a power of 2.

From $2^k - 1$ you may take at most half, which leaves something strictly between $2^{k-1} - 1$ and $2^k - 1$: never another losing pile. From any other pile you can take just enough to land on the next $2^k - 1$ below. From 300, take 45 to leave 255.

You are Bob. Alice (computer) always moves to a losing position (orange). Number line: orange = L-positions ($2^k - 1$).



move to (128–254):

Alice starts: 300 → 255.

Your move from 255. Every move lands on a non-orange number.

Solution

Form 5 (game) · **Level** = marbles in the pile n .

$\text{win}(n)$ is about the player about to move. No legal move $\rightarrow$ lose (base case).

1. **LEVEL.** $\text{win}(n) = \text{true}$ if the player about to move with n marbles can force a win. The answer is $\text{win}(300)$.

Why: the pile size is the whole configuration.

2. **CHOICE.** Remove r marbles.

Why: the rules.

3. **CHECK.** $1 \leq r \leq \lfloor n/2 \rfloor$. Base case: no legal r ($n = 1$) $\rightarrow$ lose.

Why: $\lfloor 1/2 \rfloor = 0$.

4. **MOVE.** $\text{win}(n) = \text{OR}$ over legal r of NOT $\text{win}(n - r)$.

Why: Form 5: you win iff some move leaves the opponent (now the player to move) in a losing position.

5. **CODE.**

```
int dp[301]; // all -1
bool win(int n) {
    if (dp[n] != -1) return dp[n]; // cache check (n = 1: lose)
    bool ans = false;
    for (int r = 1; r <= n / 2; r++) // choice + check
        if (!win(n - r)) ans = true; // move
    return dp[n] = ans; // save and return
} // win(300) cost: 300 states x 150 transitions
```

Why: his fixed order; the base case is the empty loop.

6. **PATTERN.** Printing $\text{win}(n)$: the losing n are 1, 3, 7, 15, 31, ..., i.e. $n = 2^k - 1$.

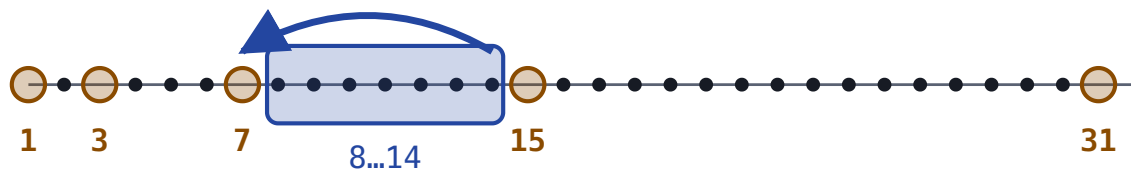
Why: run the code for $n \leq 40$.

7. **PROVE (i): every move from $2^k - 1$ goes to a winning n .** $r \leq \lfloor n/2 \rfloor = 2^{k-1} - 1$, so what is left is between $2^k - 1 - (2^{k-1} - 1) = 2^{k-1}$ and $2^k - 2$. No number there has the form $2^j - 1$.

Why: it lies strictly between $2^{k-1} - 1$ and $2^k - 1$.

8. **PROVE (ii): from any other $n \geq 2$ there is a move to a losing n .** Pick k with $2^{k-1} \leq n \leq 2^k - 2$ and remove $r = n - (2^{k-1} - 1) \geq 1$. It is legal because $r \leq n/2 \Leftrightarrow n \leq 2^k - 2$. It leaves $2^{k-1} - 1$.

Why: (i) and (ii) are exactly the two properties that define L and W , so $L = \{2^k - 1\}$.



$L = 2^k - 1$ (orange); 8...14 can all drop to 7
but from 15 no move reaches 7

9. **SOLVE.** 300 is not of that form ($256 \leq 300 \leq 510$, so $k = 9$): Alice removes $300 - 255 = 45$, leaving $255 = 2^8 - 1$.

Why: (ii) with $n = 300$.

Answer. Alice wins: take 45, leaving 255, then always return Bob to the next $2^k - 1$. ■

△ Course text has wrong bounds ("between $2^k - 1$ and 2^{k-2} ") and doesn't check $r \leq n/2$ (PROVE (ii)).

Divisors II

The number 60 is written on a board. Alice and Bob take turns, Alice first. A move: if the number on the board is n , pick any divisor d of n (1 and n itself are allowed) and replace n by $n - d$. The player who writes 0 loses.

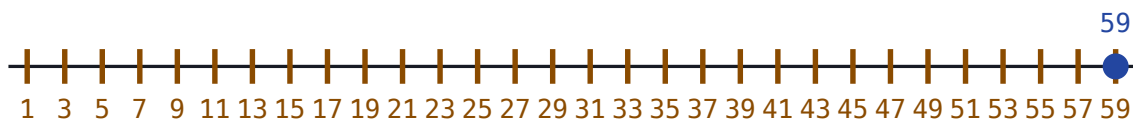
Question: which player wins?

Picture

Odd numbers lose; even numbers win.

An odd number has only odd divisors, and odd - odd is even: from odd you must hand your opponent an even number. From even you subtract 1 and hand back an odd number. From 60, write 59.

You are Bob. Alice (computer) always moves to a losing position (orange). Number line: orange = L-positions (odd numbers).



→ 58

Restart

Alice starts: 60 → 59.

Your move from 59. Every move lands on a non-orange number.

Solution

Form 5 (game) · **Level** = the number n on the board. $\text{win}(n)$ is about the player about to move. Writing 0 loses.

1. **LEVEL.** $\text{win}(n) = \text{true}$ if the player about to move with n on the board ($n \geq 1$) can force a win. The answer is $\text{win}(60)$.

Why: the number on the board is the whole configuration.

2. **CHOICE.** A divisor d of n (1 and n allowed); write $n - d$.

Why: the rules.

3. **CHECK.** $d = n$ writes 0 : the mover loses at once, so it is never a winning choice. Base case: $n = 1 \rightarrow$ the only divisor is 1 , which writes $0 \rightarrow$ lose.

Why: the rules.

4. **MOVE.** $\text{win}(n) = \text{OR over divisors } d < n \text{ of NOT win}(n - d)$.

Why: Form 5: win iff some move leaves the opponent in a losing position.

5. **CODE.**

```
int dp[61]; // all -1
bool win(ll n) {
    if (dp[n] != -1) return dp[n]; // cache check (n = 1: no)
    bool ans = false;
    for (ll d = 1; d < n; d++) // choice
        if (n % d == 0 && !win(n - d)) ans = true; // check, move
    return dp[n] = ans; // save and return
} // win(60)
```

Why: his fixed order; the base case is the empty loop.

6. **PATTERN.** Printing $\text{win}(n)$: the losing n are $1, 3, 5, 7, \dots$: exactly the odd numbers.

Why: run the code for $n \leq 20$.

7. **PROVE.** Strong induction on n .

$n = 1$: lose (CHECK).

Odd $n > 1$: every divisor d is odd, so $n - d$ is even. $d = n$ loses at once, and $d < n$ gives an even

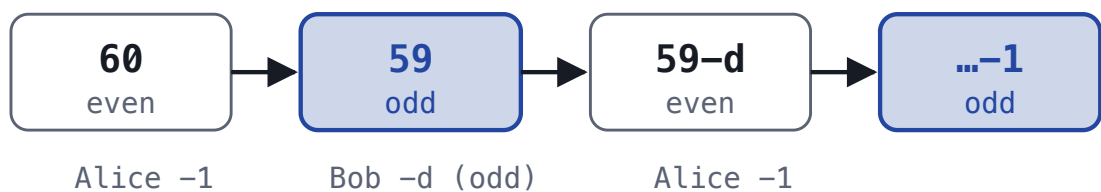
$n - d \geq 2$, which is winning by induction. **So odd n is losing.**

Even n : $d = 1$ gives $n - 1$, odd, losing by induction. **So even n is winning.**

Why: odd numbers have only odd divisors, and odd - odd is even.

8. **SOLVE.** 60 is even: Alice writes **60 - 1 = 59**, and keeps handing Bob odd numbers.

Why: PROVE with $n = 60$.



odd $\Rightarrow$ only odd divisors $\Rightarrow$ Bob always gets odd

Answer. Alice wins $(60 \rightarrow 59)$. ■

4 · Endgame analysis

Work backwards from the final position and label each position W or L. Then read off the pattern and prove it as in §3.

Moving king

A king stands in the bottom-left corner of an 8×8 board. Alice and Bob take turns moving it, Alice first. A move is one square up, one square right, or

one square diagonally up-and-right. The player who moves the king into the top-right corner wins.

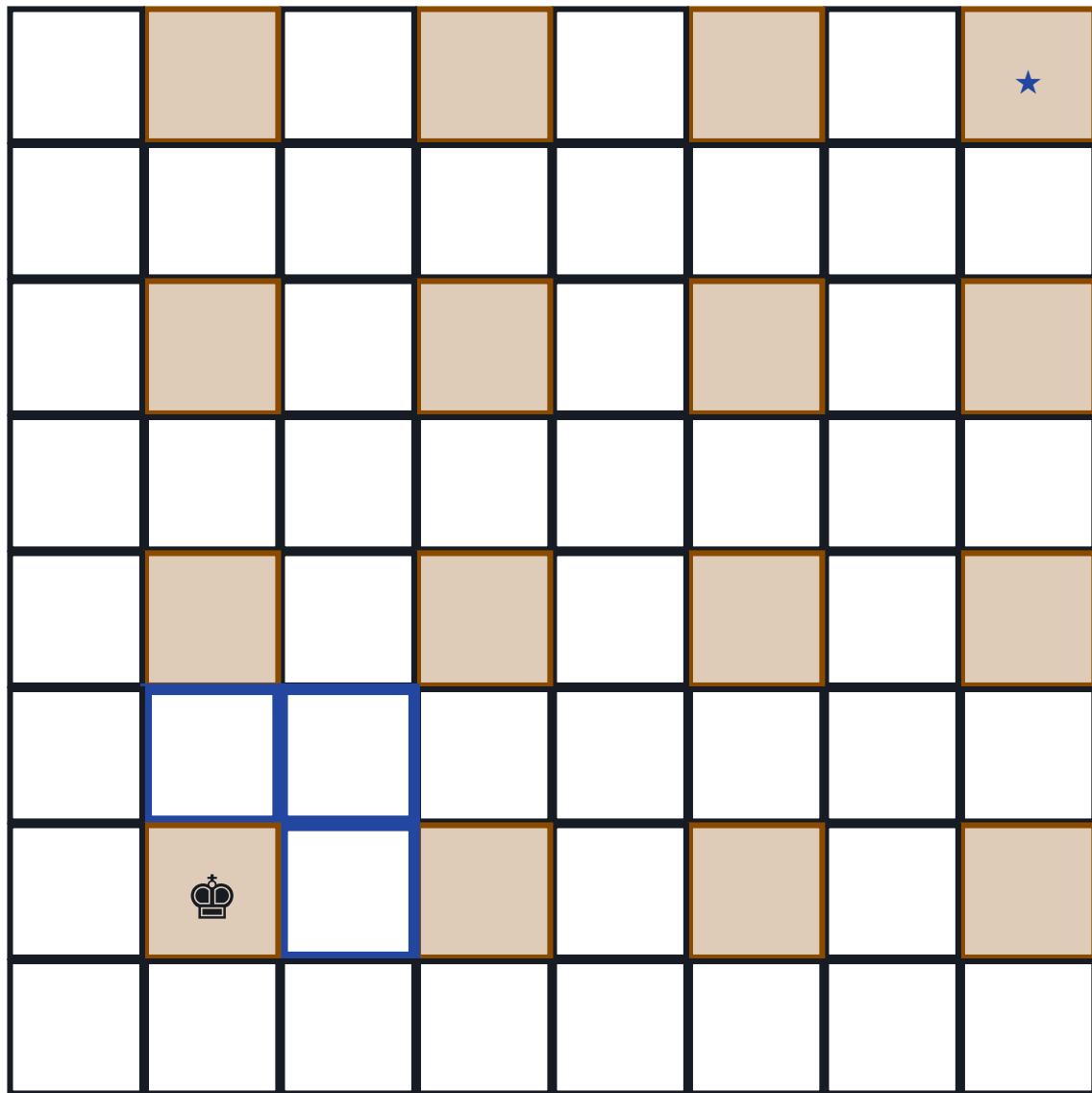
Question: which player wins?

Picture

Losing squares: both distances to the corner even.

Every king move changes one or both distances by 1, so from (even, even) you always leave something odd. From anything else you can fix the odd ones in one move and return to (even, even). Alice starts on (7, 7) and moves diagonally to (6, 6).

Orange squares are losing for the player about to move (both distances even). You are Bob: click a square the king can move to. Alice always returns to orange.



Alice opened with the diagonal move to (6, 6), an orange square. Your move.

[Restart](#)

Solution

Form 5 (game) · **Level** = the distance still to go (a up, b right). `win(a, b)` is about the player about to move. The player who reaches (0, 0) wins.

1. **LEVEL.** `win(a, b) = true` if the player about to move, with the king still a squares below and b squares left of the top-right corner, can force a win. The start is (7, 7). The answer is `win(7, 7)`.

Why: the king's position is the whole configuration.

2. **CHOICE.** Up $(a-1, b)$, right $(a, b-1)$, or diagonal $(a-1, b-1)$.

Why: the rules.

3. **CHECK.** Stay ≥ 0 in both. Base case: $\text{win}(0, 0) = \text{false}$, because whoever faces $(0, 0)$ has just watched the opponent reach the corner.

Why: reaching the corner wins.

4. **MOVE.** $\text{win}(a, b) = \text{OR over legal moves of NOT win(next)}$.

Why: Form 5.

5. CODE.

```
int dp[8][8]; // all -1
bool win(ll a, ll b) {
    if (a == 0 && b == 0) return false; // base case: opponent ju
    if (dp[a][b] != -1) return dp[a][b]; // cache check
    bool ans = false;
    if (a >= 1 && !win(a - 1, b)) ans = true; // up
    if (b >= 1 && !win(a, b - 1)) ans = true; // right
    if (a >= 1 && b >= 1 && !win(a - 1, b - 1)) ans = true; // diagonal
    return dp[a][b] = ans; // save and return
} // win(7, 7) cost: 64 states x 3 transitions
```

Why: his fixed order.

6. **PATTERN.** The table of $\text{win}(a, b)$ (rows a , columns $b = 0 \dots 7$):

a \ b	0	1	2	3	4	5	6	7
7	W	W	W	W	W	W	W	W
6	L	W	L	W	L	W	L	W
5	W	W	W	W	W	W	W	W
4	L	W	L	W	L	W	L	W
3	W	W	W	W	W	W	W	W

2	L	W	L	W	L	W	L	W
1	W	W	W	W	W	W	W	W
0	L	W	L	W	L	W	L	W

L exactly when a and b are both even.

Why: run the code.

							★
K							

shaded = L (both distances even)

7. **PROVE.** From (even, even) $\neq$ (0, 0) every move makes a or b odd (each move subtracts 1 from a, from b, or from both). From any other position you can reach (even, even): if both are odd move diagonally, if one is odd reduce that one.

Why: subtracting 1 flips parity; with the base (0, 0) these are the two properties that define L and W.

8. **SOLVE.** (7, 7) is not (even, even), so Alice wins: she moves **diagonally to (6, 6)** and always restores (even, even).

Why: PROVE.

Answer. Alice wins. ■

5 · Ad-hoc

Look for a quantity fixed by the rules (so the total number of moves is decided in advance), or a pairing/colouring from the brainteasers toolkit.

Three piles

There are three piles with 10, 15 and 20 stones. Alice and Bob take turns, Alice first. A move: choose a pile with at least 2 stones and split it into two non-empty piles. The player who cannot move loses.

Question: which player wins?

Key idea

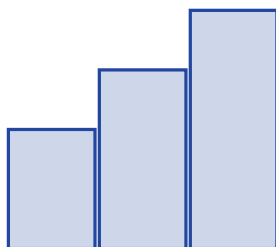
Each move adds exactly one pile. So the number of moves is fixed.

Picture

Every move makes exactly one more pile, so the length of the game is fixed.

It starts with 3 piles and ends with 45 single stones. That is 42 moves whatever anyone does. An even number of moves means Bob makes the last one.

Click any pile of 2 or more to split it at a random point. Try to change the number of moves: you can't.



Random split

Play to the end

Restart

Moves so far **0**. Piles **3**. Moves still to come: $45 - 3 = \mathbf{42}$, whatever anyone does.

Solution

Setup. Piles of 10, 15 and 20 stones. A move splits one pile of at least 2 stones into two non-empty piles. The player who can't move loses. Alice moves first.

Claim. Bob wins.

1. Each move adds exactly 1 pile.

Why: one pile becomes two.

3 piles

10

15

20

4 piles

10

7

8

20

every move: +1 pile. End: 45 piles of 1
 $\text{moves} = 45 - 3 = 42$ (even) $\Rightarrow$ Bob moves last

2. The game ends only when every pile has 1 stone, i.e. when there are $10 + 15 + 20 = 45$ piles.

Why: a pile of ≥ 2 can still be split.

3. So exactly $45 - 3 = 42$ moves are played, whatever anyone does.

Why: steps 1–2.

4. 42 is even, so the last move is Bob's, and Alice is then stuck.

Why: moves alternate starting with Alice.

Conclusion. The number of moves is fixed at 42, so Bob moves last. ■

Moving the piece

A board has 183 rows and 181 columns. A piece starts in the top-left square, which counts as visited. Alice and Bob take turns moving the piece, Alice first. A move takes the piece to a neighbouring square (sideways or diagonally) that has not been visited yet. The player who cannot move loses.

Question: which player wins?

Key idea

mn is odd, so the cells other than the start can be split into adjacent pairs. Bob always moves to the partner of Alice's cell.

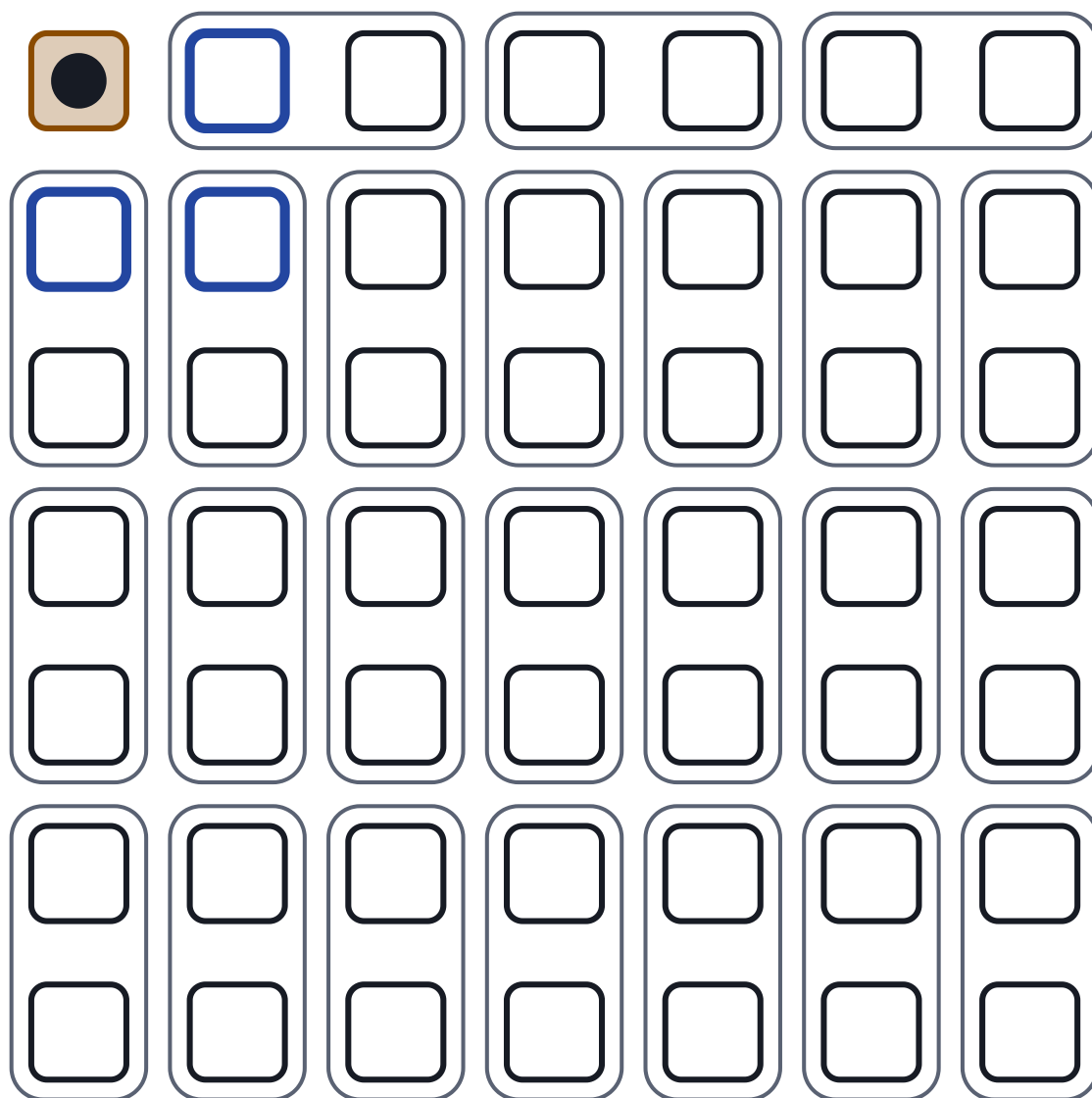
Picture

Cover every cell except the start with dominoes (possible because the board is odd $\times$ odd).

Whenever Alice steps onto a domino, Bob steps onto its other half. So Alice always needs a fresh domino, and she is the one who runs out.

A 7×7 board (odd $\times$ odd, same pairing). Dominoes pair every cell except the start. You are Alice: click a neighbouring unvisited

cell. Bob always moves to the other half of your domino.



The piece starts in the top-left corner. Your move.

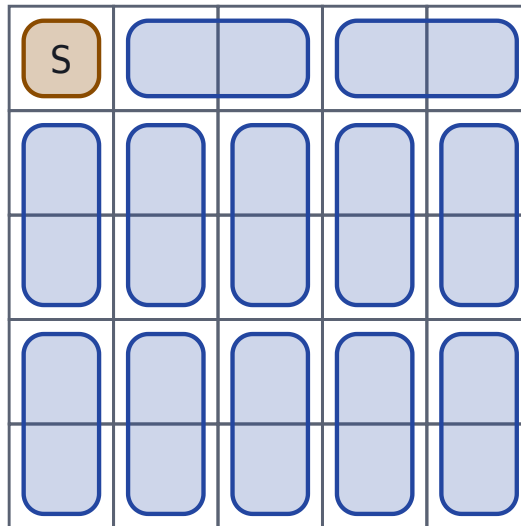
[Restart](#)

Solution

Setup. A 183×181 grid (183 rows, 181 columns). The piece starts in the top-left cell, which counts as visited. A move goes to an adjacent unvisited cell (diagonals count). The player who can't move loses. Alice moves first.

Claim. Bob wins.

1. Pair up all cells except the start: the top row without the corner has 180 cells, paired horizontally; the other 182 rows are paired vertically, column by column (each column has 182 cells there). Partners are adjacent. Same shape on 5×5 :



Why: 180 and 182 are even, so everything pairs up.

2. Bob's rule: whenever Alice moves into a cell, Bob moves into its partner.

Why: this is the strategy.

3. Invariant: after each of Bob's moves, every pair is either fully visited or fully unvisited.

Why: true at the start (the start cell is in no pair), and each round visits exactly one whole pair.

4. So Alice always steps into an untouched pair. Its partner is adjacent and unvisited, so Bob can move there and keep the invariant.

Why: step 3.

5. Bob always has a move, and the cells run out, so Alice is stuck.

Why: finite board.

Conclusion. The pairing lets Bob answer every move.

■

△ Course shows only a coloured 5×5 picture. Step 1 is the explicit pairing for any odd m, n .